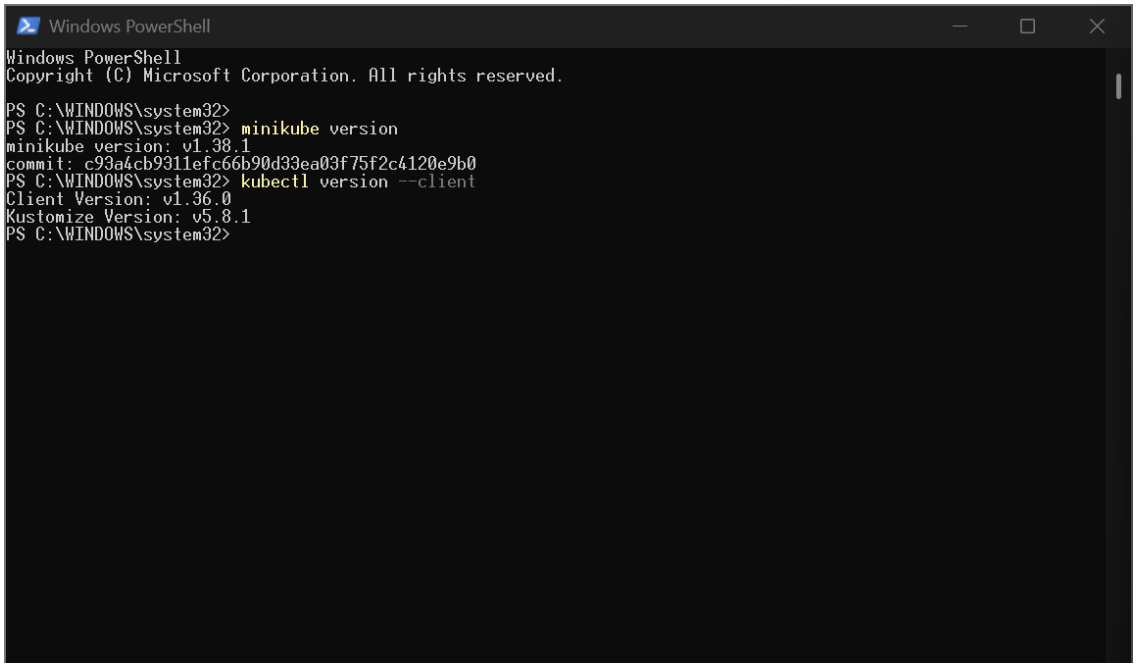
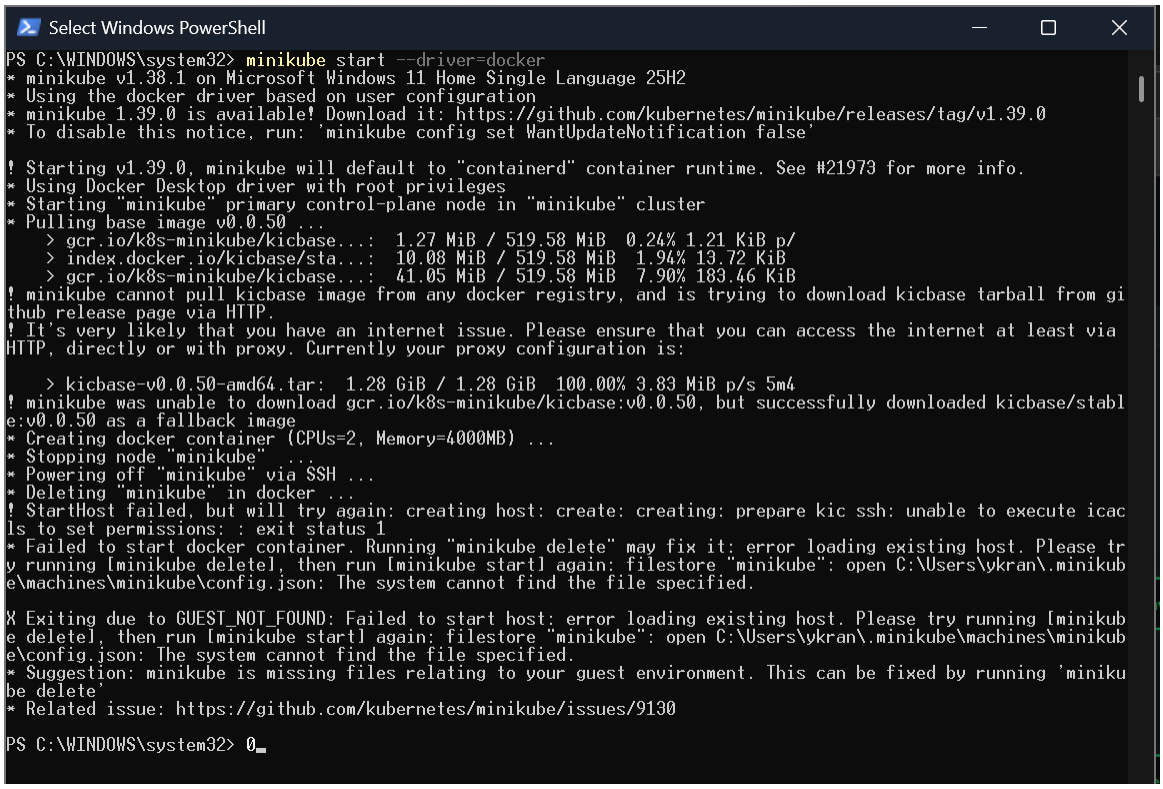
Roll: 2420090043
Name: R. Rahul

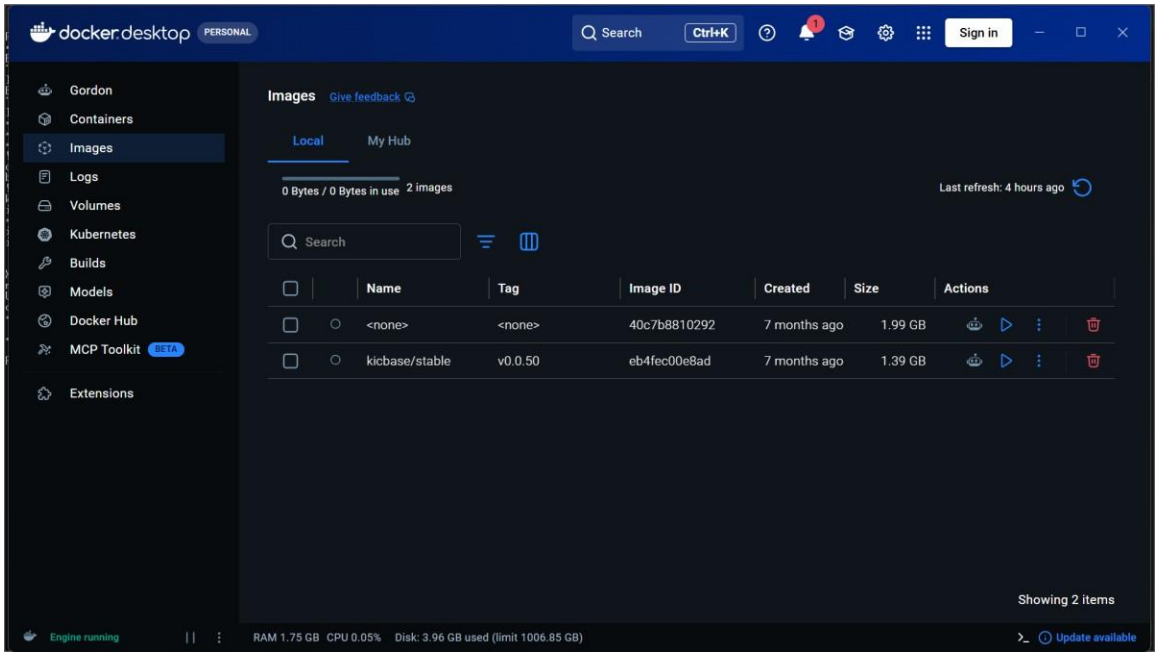
Kubernetes

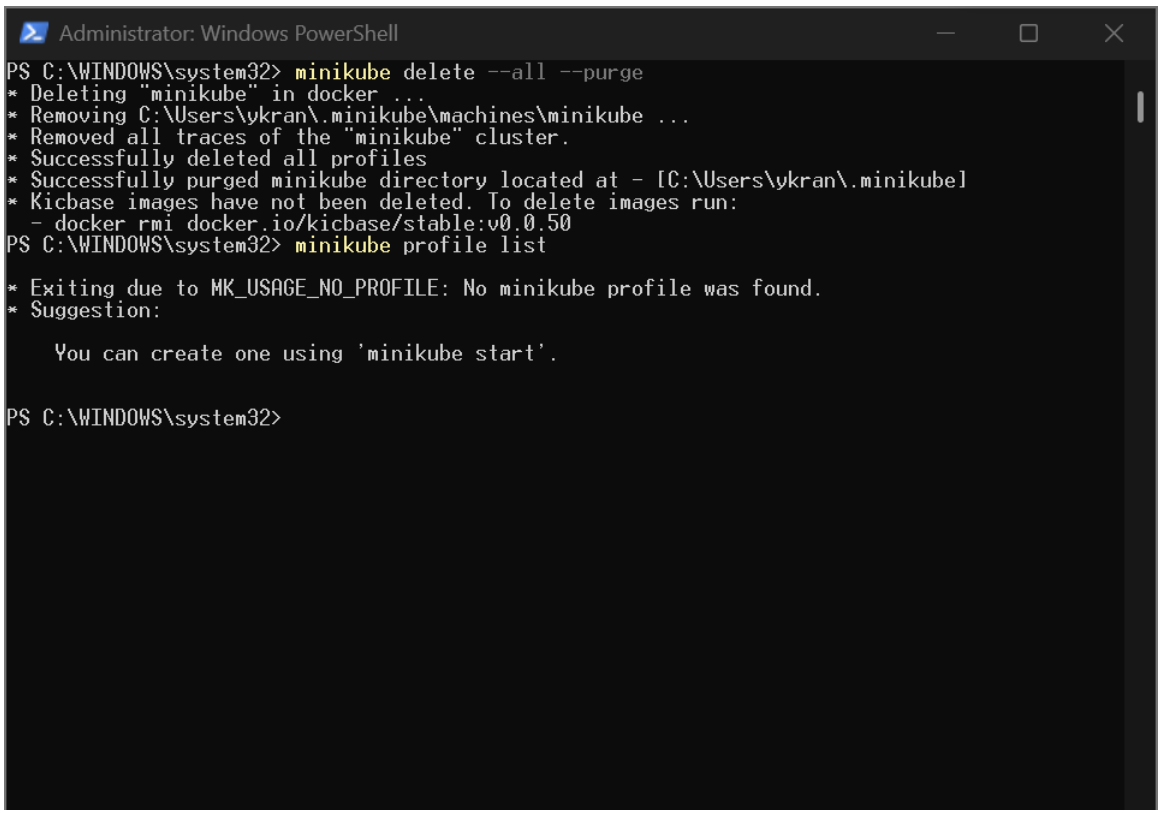
1. Install **Docker Desktop** and make sure it is running.
2. Open **PowerShell as Administrator**.
3. Install **Minikube** using:
   winget install Kubernetes.minikube
4. Check Minikube using:
   minikube version
5. Install **kubectl** using:
   winget install -e --id Kubernetes.kubectl
6. Check kubectl using:
   kubectl version --client

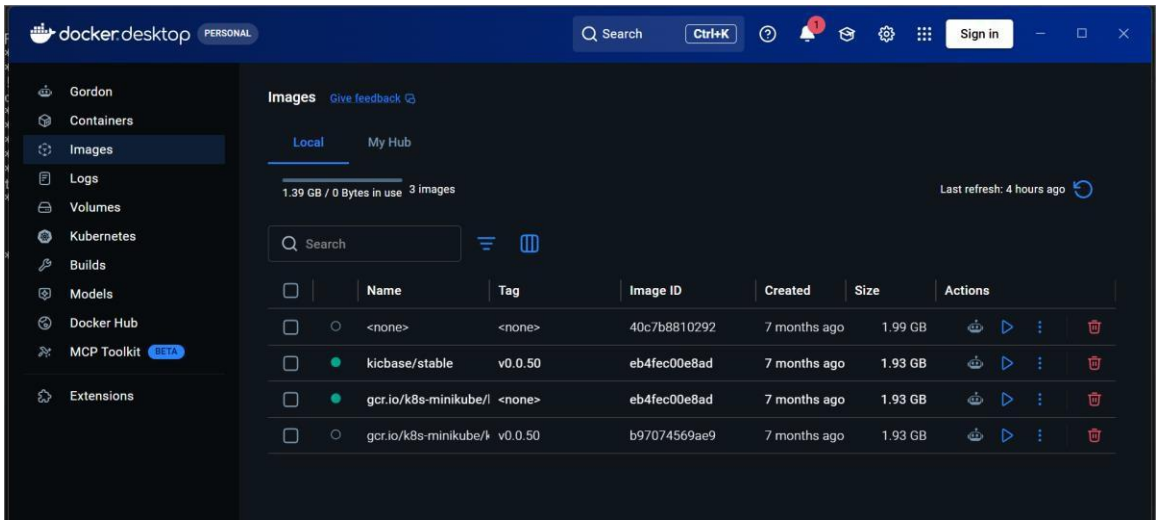


7. Start Minikube using:
   minikube start --driver=docker

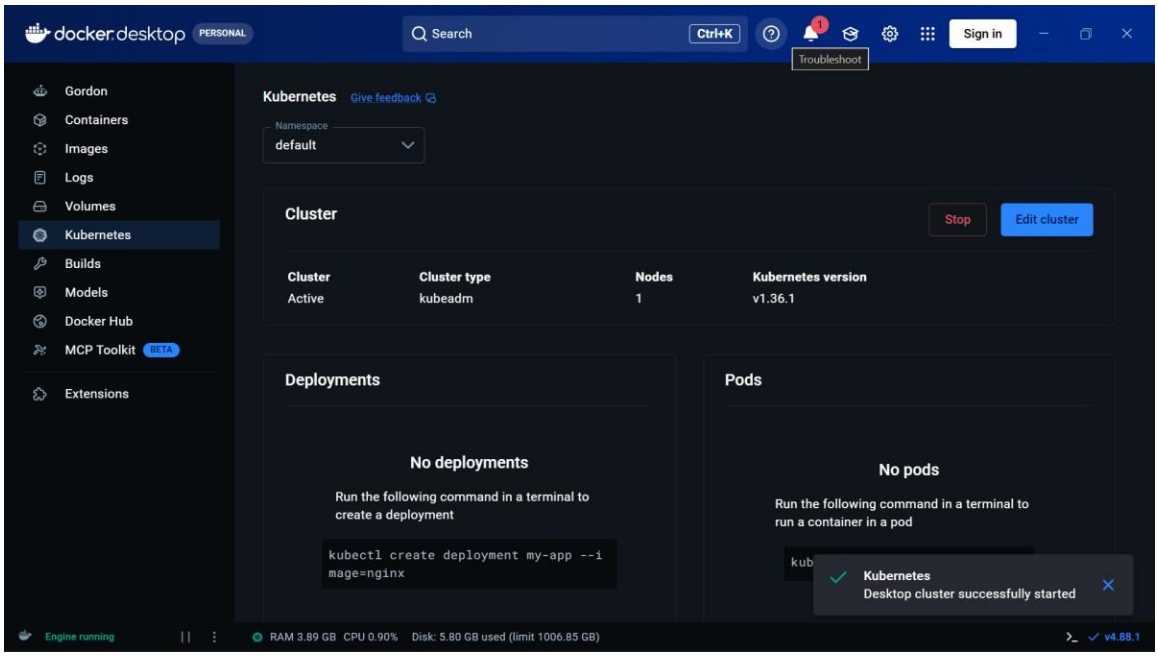






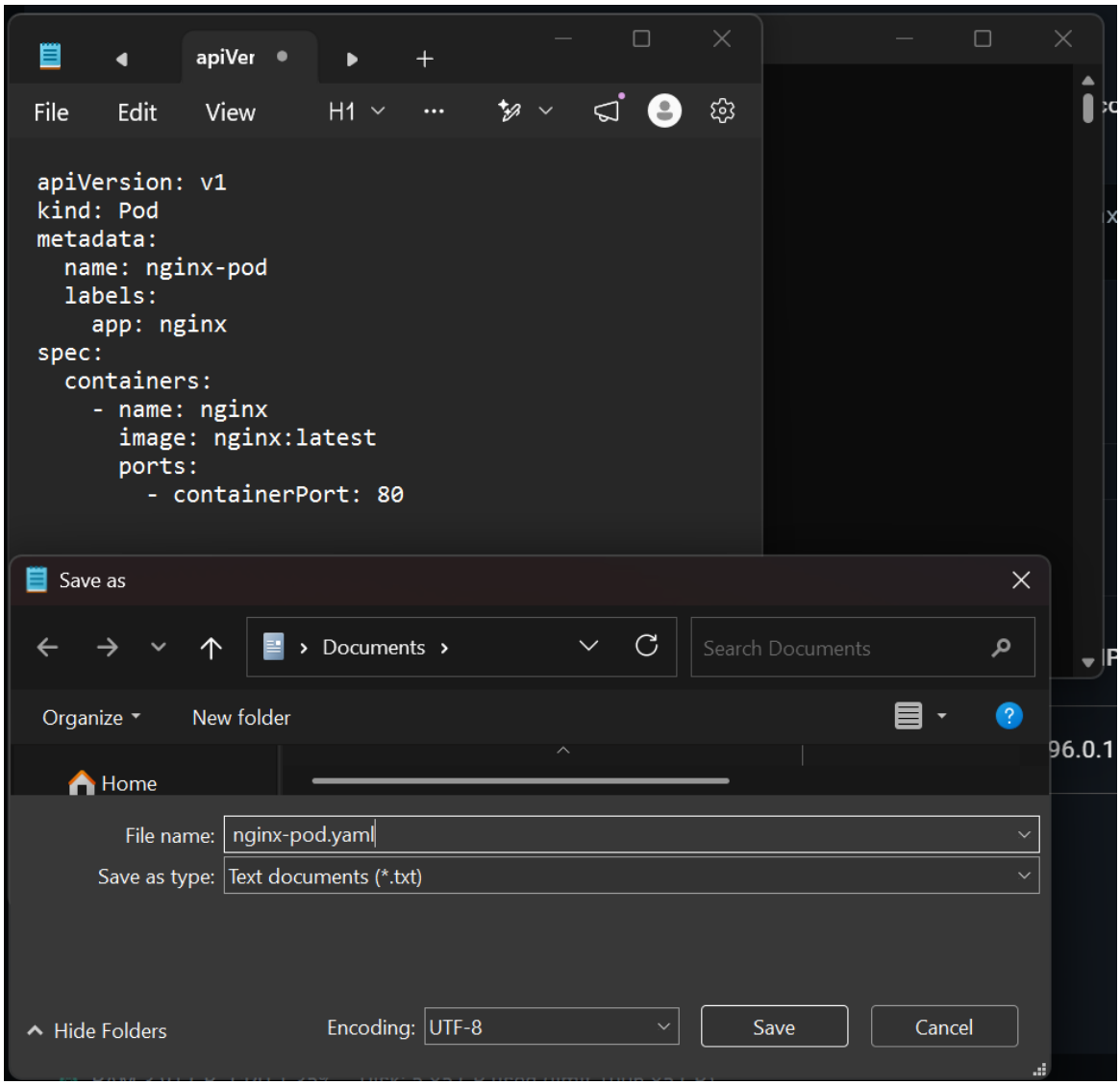


8. Check the cluster status using:
   minikube status

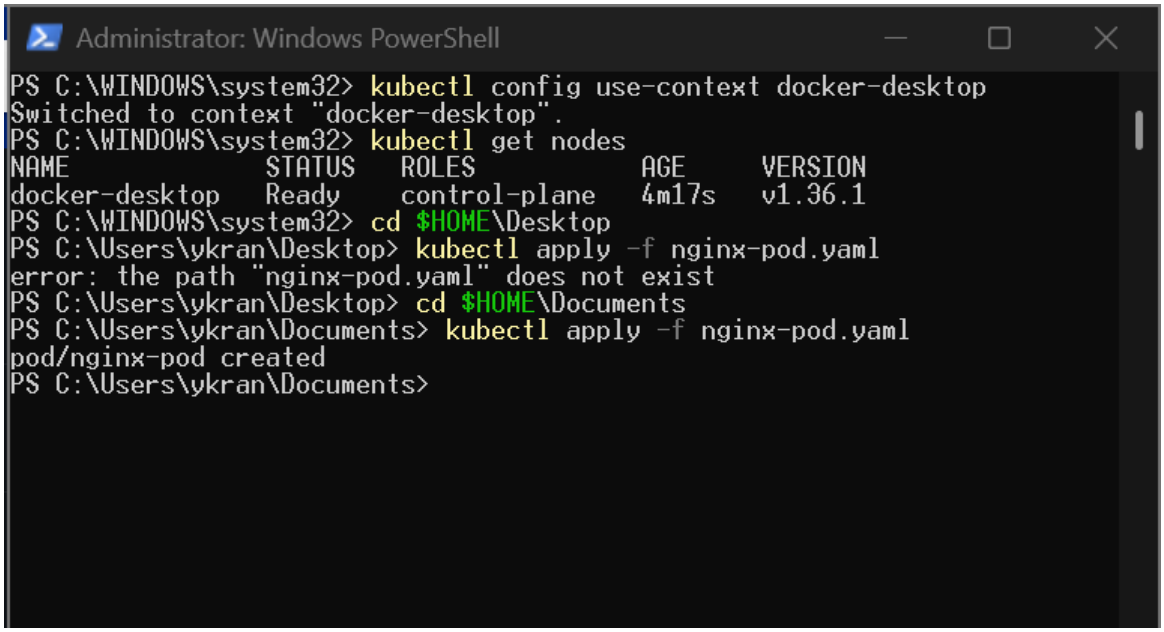


9. Check the Kubernetes node using:
   kubectl get nodes

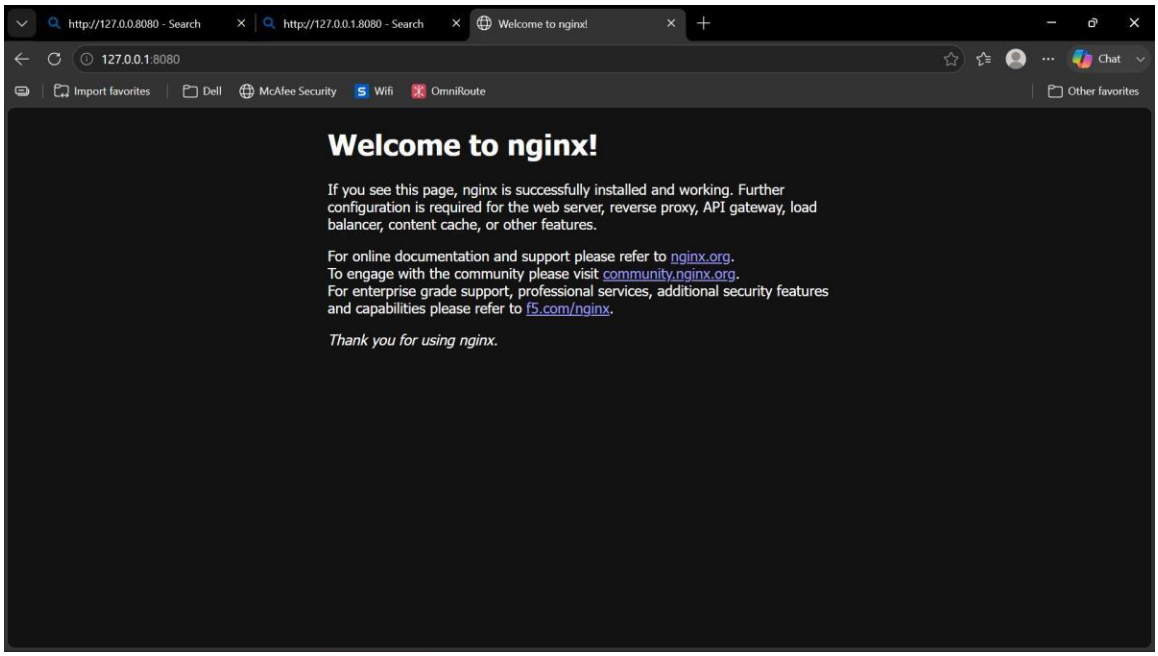
10. Create a file named **nginx-pod.yaml** and add the Nginx Pod configuration.



11. Deploy the Nginx Pod using:
kubectl apply -f nginx-pod.yaml



12. Check the Pod status using:
    kubectl get pods
13. View Pod details using:
    kubectl describe pod nginx-pod
14. Access Nginx using:
    kubectl port-forward pod/nginx-pod 8080:80
15. Open http://localhost:8080 in the browser.
16. Verify that the **Welcome to nginx!** page is displayed.



17. View Nginx logs using:
    kubectl logs nginx-pod

18. Delete the Pod using:
    kubectl delete -f nginx-pod.yaml

19. Stop Minikube using:
    minikube stop